



EEE 4554 Project Final Report

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Project P17: Power Amplifier (PA) Nonlinearity and Digital Predistortion (DPD)

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1. Introduction

Power amplifiers are important components in wireless communication systems because they increase the signal power before transmission. However, when a power amplifier operates close to its high-power region, it does not behave perfectly linearly. This nonlinear behavior distorts the transmitted signal and creates unwanted spectral components outside the desired frequency band. These nonlinear effects can reduce modulation quality and cause spectral regrowth, which may interfere with adjacent channels. Therefore, improving power amplifier linearity is an important problem in RF communication systems.

Digital predistortion is a commonly used technique to reduce power amplifier nonlinearity. The main idea is to modify the input signal before the power amplifier so that the distortion introduced by the amplifier is compensated at the output.

In this project, a simulation-based study is carried out to analyze power amplifier nonlinearity and digital predistortion. A pulse-shaped 16-QAM signal is generated and passed through a memoryless nonlinear power amplifier model. First, the system is evaluated without digital predistortion. Then, two predistortion approaches are implemented and compared: a polynomial DPD method and a LUT-based DPD method.

The performance is evaluated using NMSE, EVM, ACPR, PSD, and constellation diagrams. In addition, an IBO-based comparison and a generalization test are performed to observe how the DPD models behave under different operating conditions. The aim is to compare the baseline nonlinear PA output with the improved outputs obtained using polynomial and LUT-based digital predistortion.

2. System Model

The system model used in this project is based on a baseband simulation chain. First, a 16-QAM signal is generated and pulse-shaped using a root raised cosine filter. Then, the signal is passed through a nonlinear power amplifier model. The amplifier output is evaluated in three cases: without DPD, with polynomial DPD, and with LUT-based DPD.

For the baseline case, the predistortion block is bypassed and the signal is directly applied to the nonlinear PA. For the improved cases, the input signal is first modified by a predistorter before entering the PA. The main purpose of this system model is to observe the effect of PA nonlinearity and compare how different DPD methods reduce the distortion.

2.1 Modulated Signal Generation

In this study, a 16-QAM signal is used as the input waveform. 16-QAM is selected because it has amplitude variations, which makes nonlinear distortion easier to observe compared to constant-envelope modulations such as BPSK or QPSK. At the same time, it is less complex than higher-order modulation schemes such as 64-QAM.

After symbol generation, a root raised cosine (RRC) filter is used for pulse shaping. The RRC filter reduces intersymbol interference and controls the signal bandwidth. This makes the generated waveform closer to a practical digital communication signal.

The MATLAB function `generate_modulated_waveform()` is used to generate random 16-QAM symbols, apply RRC pulse shaping, and remove the filter delay from the output waveform.

2.2 Power Amplifier Model

The power amplifier is modeled using a memoryless fifth-order odd polynomial model. This model is selected as a simple baseline to represent amplitude-dependent nonlinear distortion in RF power amplifiers. Since the model is memoryless, the current output depends only on the current input sample.

The PA output is expressed as

$$y(n) = a_1 x_{in}(n) + a_3 |x_{in}(n)|^2 x_{in}(n) + a_5 |x_{in}(n)|^4 x_{in}(n)$$

where $x_{in}(n)$ is the amplifier input after input backoff scaling, and a_1 , a_3 , and a_5 are the polynomial coefficients.

Before the nonlinear PA model is applied, the input signal is scaled according to the input backoff:

$$x_{in}(n) = x(n) 10^{-IBO_{dB}/20}$$

In this project, the PA coefficients are selected as

$$a_1 = 1.0, a_3 = -0.15, a_5 = -0.05$$

The negative higher-order coefficients produce gain compression at larger input amplitudes. Therefore, nonlinear distortion becomes more visible at low IBO values, while the PA behaves more linearly at higher IBO values.

Since only real-valued coefficients are used, the model mainly represents AM/AM distortion and does not explicitly include AM/PM effects. If a complex Saleh-type model were used, the output phase would also change with input amplitude, and the DPD model would need to compensate both amplitude and phase distortion.

2.3 Polynomial Digital Predistortion

Polynomial DPD is used as the first predistortion method. The basic idea is to modify the input signal before the PA so that the nonlinear distortion introduced by the PA is reduced at the output. The predistorted signal is modeled as

$$x_{pre}(n) = x(n) + c_3 |x(n)|^2 x(n)$$

where c_3 is the predistortion coefficient. This coefficient controls the strength of the nonlinear correction.

The coefficient c_3 is learned using a grid-search-based training procedure. Candidate c_3 values are tested over a predefined range. For each candidate value, the input signal is predistorted, passed through the PA model, and compared with an ideal linear reference signal. The value that minimizes the NMSE is selected as the optimal coefficient.

The optimization objective is

$$c_3^* = \arg \min_{c_3} \frac{\sum_n |y(n) - x_{ref}(n)|^2}{\sum_n |x_{ref}(n)|^2}$$

where $y(n)$ is the PA output after predistortion and $x_{ref}(n)$ is the ideal linear reference signal.

This polynomial DPD model is simple, interpretable, and computationally light. However, it mainly targets the dominant third-order nonlinear effect and may be limited for more complex PA behavior.

2.4 LUT-Based Digital Predistortion

As an extension method, a lookup table (LUT)-based DPD model is implemented. Unlike the polynomial DPD model, LUT-based DPD does not assume a fixed polynomial form. Instead, it learns amplitude-dependent correction values from the training signal.

In the LUT-based method, the input amplitude range is divided into several bins. For each amplitude bin, a correction factor is estimated by comparing the PA output with the ideal linear reference. During testing, the input sample amplitude is used to select the corresponding correction value from the lookup table.

The main idea can be summarized as follows:

$$|x(n)| \mapsto \text{LUT} \rightarrow \text{correction factor}$$

Then, the correction factor is applied to the input signal before it enters the PA.

This approach is flexible because different amplitude regions can have different correction values. Therefore, LUT-based DPD can adapt to nonlinear behavior without explicitly defining a polynomial equation. However, its performance depends on the number of LUT bins and the quality of the training data.

In this project, LUT-based DPD is used as the comparison method against polynomial DPD. This allows the system to compare a simple analytical predistorter with a data-driven amplitude-based predistorter.

3. Simulation Setup

The simulation is implemented in MATLAB using a reproducible baseband setup. The same waveform, PA model, and performance metrics are used for all compared cases: No DPD, Polynomial DPD, and LUT-based DPD.

The main simulation parameters are summarized below:

Parameter	Value
Modulation	16-QAM
Pulse shaping filter	Root Raised Cosine (RRC)
Roll-off factor	0.25
Filter span	10 symbols
Samples per symbol (sps)	4
Sampling frequency	100 MHz

IBO values	0, 3, 6, 9 dB
PA model	Memoryless 5th-order odd polynomial
PA coefficients	($a_1=1.0$, $a_3=-0.15$, $a_5=-0.05$)
Polynomial DPD model	3rd-order polynomial predistorter
LUT-based DPD	Amplitude-bin-based correction
Training symbol count	5800
Test symbol count	2000
Random seed	3787
PSD method	Welch method
PSD FFT size	2048
Performance metrics	NMSE, EVM, ACPR, PSD, constellation

The training waveform is used to estimate the DPD parameters, while the test waveform is used only for performance evaluation. This separation is used to avoid evaluating the predistorter on the same signal used during training.

For the polynomial DPD, the coefficient c_3 is selected using grid search. For high IBO values, the c_3 search range is reduced to avoid overcompensation since the PA operates closer to the linear region.

For the LUT-based DPD, the input amplitude range is divided into bins, and a correction factor is learned for each bin using the training waveform.

The PSD is computed using Welch's method, and the system performance is evaluated using NMSE, EVM, ACPR, PSD plots, and constellation diagrams.

4. Results

The simulation results compare three cases: No DPD, Polynomial DPD, and LUT-based DPD. The results are evaluated using AM/AM characteristics, PSD, NMSE, ACPR, EVM, constellation diagrams, and a generalization test.

4.1 AM/AM Characteristics

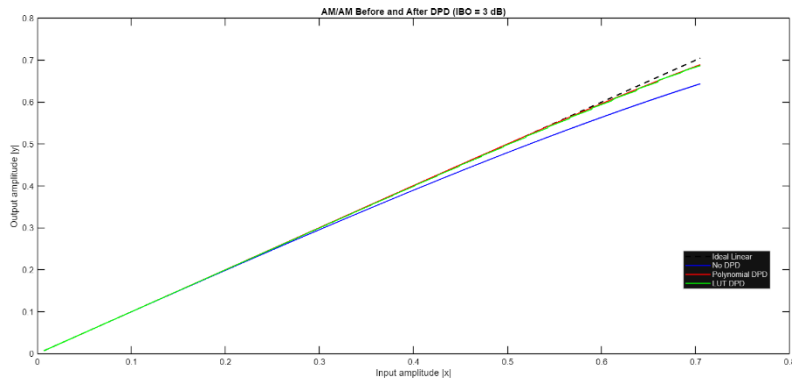


Figure 1 AM/AM characteristic at IBO = 3 dB

shows that both DPD methods reduce amplitude distortion. Polynomial DPD gives a slightly closer response to the ideal curve.

4.2 PSD and Spectral Regrowth

Figure 2 shows the PSD before and after DPD at IBO = 3 dB. Without DPD, spectral regrowth is clearly observed outside the main signal band. After applying Polynomial DPD and LUT DPD, the out-of-band components are reduced.

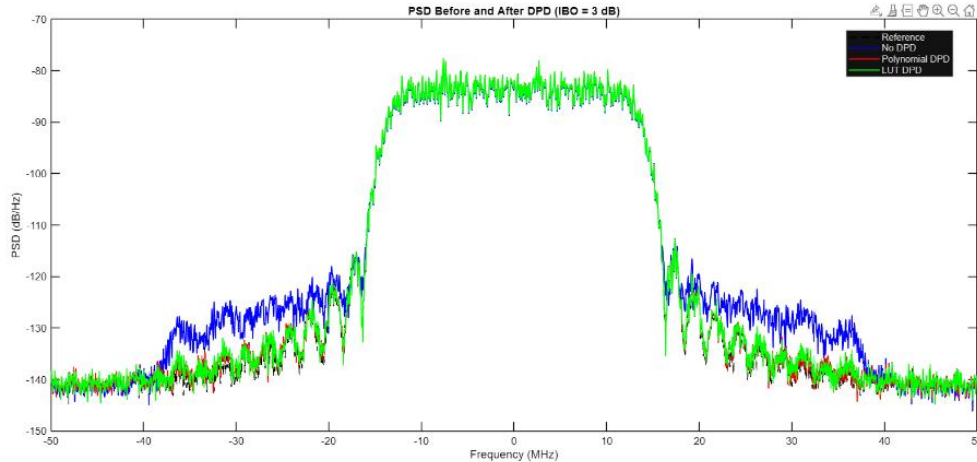


Figure 2 PSD before and after DPD at IBO = 3 dB

The PSD zoom-in around the adjacent band shows this effect more clearly. Both DPD methods reduce adjacent-channel spectral leakage compared to the No-DPD case. Polynomial DPD and LUT DPD give very similar spectral suppression performance.

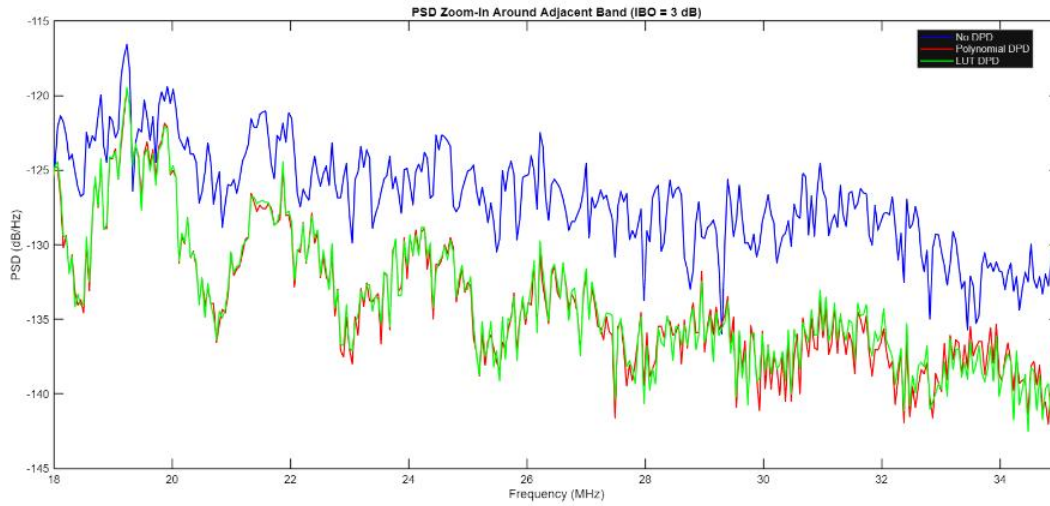


Figure 3 PSD zoom-in around the adjacent band

NMSE, ACPR, and EVM versus IBO

Figures 4 and 5 show the NMSE, ACPR, and EVM performance for different IBO values. Both Polynomial DPD and LUT DPD improve the PA output compared to the No-DPD case.

As the IBO increases, the PA operates closer to the linear region and the distortion becomes smaller. Therefore, the required predistortion strength also decreases.

Polynomial DPD generally provides the best NMSE and EVM performance, while LUT DPD gives comparable spectral regrowth suppression performance.

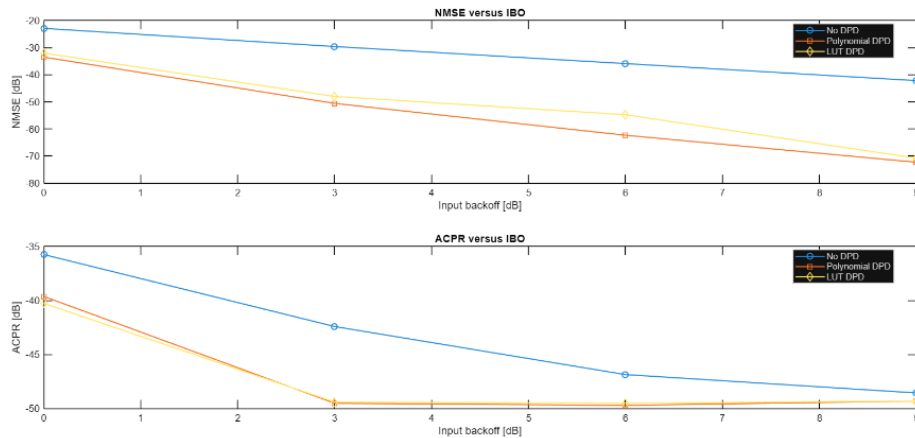


Figure 4 NMSE and ACPR comparison versus IBO

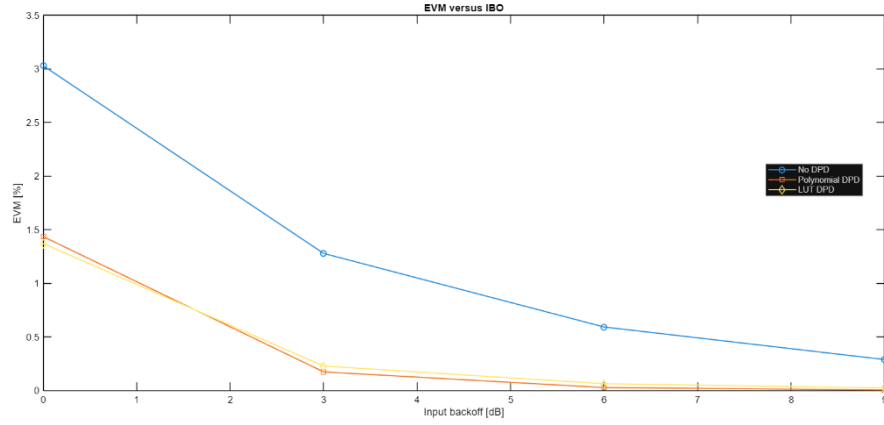


Figure 5 EVM comparison versus IBO

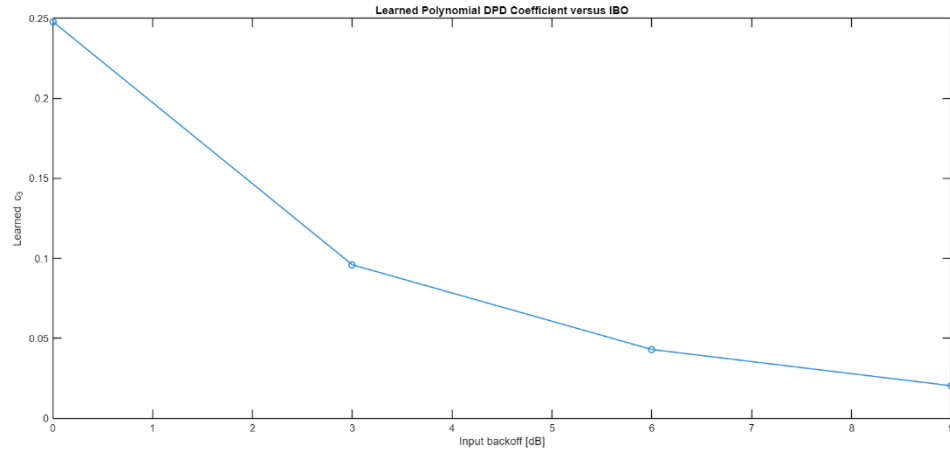


Figure 6 Learned polynomial DPD coefficient (c3) versus IBO.

Table 1 Performance Comparison of No DPD, Polynomial DPD, and LUT DPD Under Different IBO Values

IBO (dB)	NMSE No DPD (dB)	NMSE Poly DPD (dB)	NMSE LUT DPD (dB)	EVM No DPD (%)	EVM Poly DPD (%)	EVM LUT DPD (%)	ACPR No DPD (dB)	ACPR Poly DPD (dB)	ACPR LUT DPD (dB)	Learned (c_3)
0	-22.902	-33.493	-32.025	3.0296	1.4362	1.3714	-35.750	-39.621	-40.266	0.2480
3	-29.604	-50.569	-48.014	1.2791	0.1741	0.2286	-42.409	-49.512	-49.395	0.0960
6	-35.895	-62.315	-54.741	0.5929	0.0280	0.0637	-46.856	-49.699	-49.508	0.0431
9	-42.174	-72.277	-70.775	0.2896	0.0052	0.0251	-48.554	-49.305	-49.298	0.0204

Table 1 shows that both Polynomial DPD and LUT DPD improve the PA output compared to the No-DPD case for all IBO values. The NMSE values become significantly lower after applying DPD, which indicates that the PA output becomes closer to the ideal linear reference. The EVM

results also show a clear improvement in modulation accuracy. For example, at IBO = 3 dB, EVM decreases from 1.2791% to 0.1741% with Polynomial DPD and to 0.2286% with LUT DPD.

The ACPR results show that both DPD methods reduce spectral regrowth. The improvement is more visible at low and moderate IBO values, where the PA operates more nonlinearly. At higher IBO values, the PA is already closer to the linear region, so the additional ACPR improvement becomes smaller.

Overall, Polynomial DPD gives the best NMSE and EVM performance when it is trained and tested at the same IBO value. LUT DPD also provides significant improvement compared to No DPD and performs close to Polynomial DPD, especially in spectral regrowth reduction. The learned c_3 coefficient decreases as IBO increases, showing that less predistortion is required when the PA operates closer to its linear region.

4.4 Constellation Analysis

The constellation diagrams at IBO = 3 dB visually support the EVM results. Without DPD, the received symbols are more scattered around the ideal constellation points. After applying DPD, the symbol clusters become tighter.

Polynomial DPD produces the most compact clusters, which is consistent with its lower EVM value. LUT DPD also improves the constellation compared to the No-DPD case, although its EVM is slightly higher than Polynomial DPD in this test.

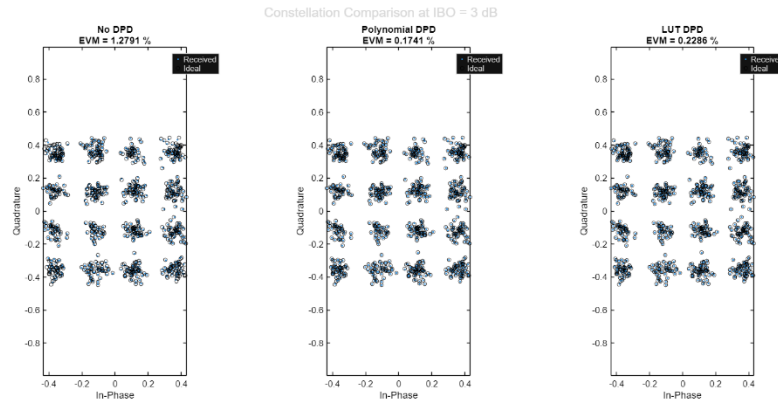


Figure 7 Constellation comparison at IBO = 3 dB for No DPD, Polynomial DPD, and LUT DPD.

4.5 Generalization Test

A generalization test is performed by training both DPD methods only at IBO = 3 dB and testing them at different IBO values. The best performance is obtained at IBO = 3 dB, which corresponds to the same operating point used during training.

When the test IBO changes to 0 dB, 6 dB, or 9 dB, the performance becomes worse for both DPD methods. This behavior is expected because the nonlinear characteristics of the power amplifier vary with the operating point. At low IBO values, the PA operates deeper in the nonlinear region, while at high IBO values the PA becomes more linear. Therefore, a predistorter optimized for one operating point cannot perfectly compensate the PA under different operating conditions.

Figures 8 and 9 show the EVM and NMSE results obtained during the generalization test. Both metrics achieve their best values at the training condition ($\text{IBO} = 3 \text{ dB}$). As the operating point moves away from the training IBO, the mismatch between the trained DPD model and the actual PA behavior increases, leading to performance degradation.

Therefore, the degradation in the generalization test does not indicate a failure of the DPD method; instead, it shows that a DPD model trained at one operating point cannot optimally compensate the PA at all IBO values.

The Polynomial DPD provides the best performance when trained and tested at the same IBO value because its coefficient is directly optimized for that specific nonlinear condition. However, LUT-based DPD performs slightly better than Polynomial DPD at higher test IBO values. This suggests that LUT-based correction may provide better robustness against operating-point variations due to its amplitude-dependent nonlinear mapping structure.

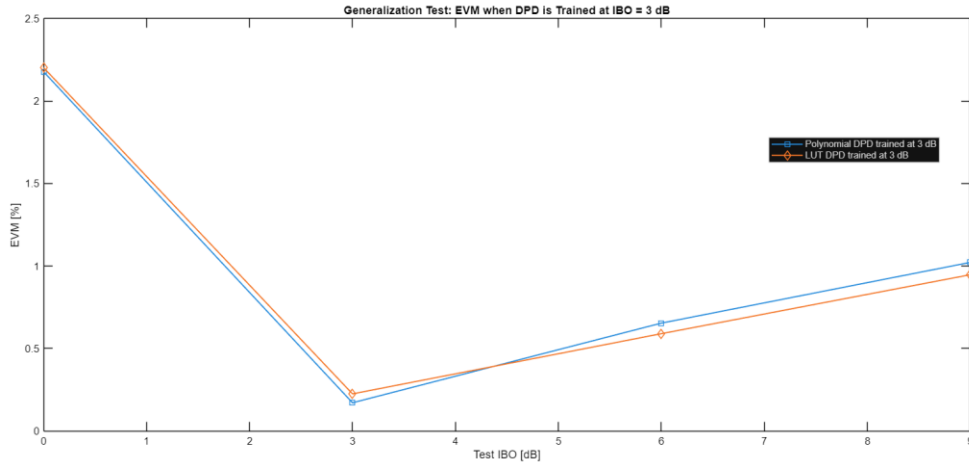


Figure 8 Generalization test EVM results when both DPD methods are trained at $\text{IBO} = 3 \text{ dB}$.

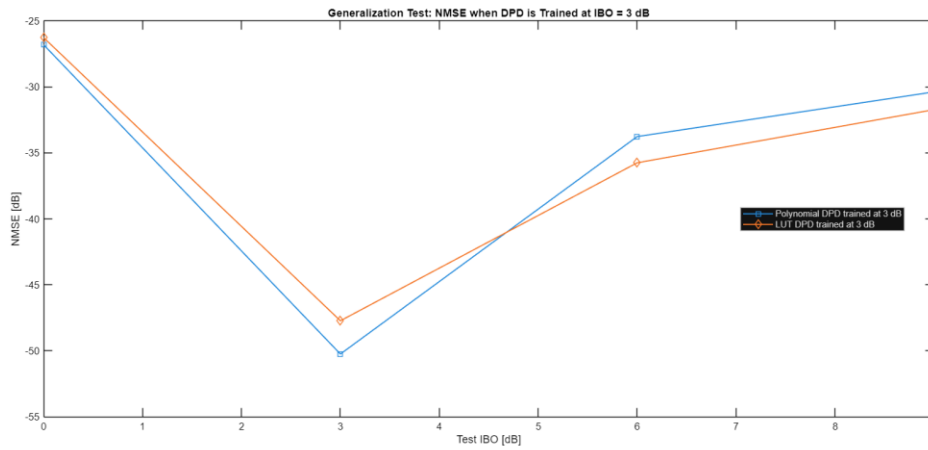


Figure 9 Generalization test NMSE results when both DPD methods are trained at $\text{IBO} = 3 \text{ dB}$.

5. Discussion

The results show that both Polynomial DPD and LUT-based DPD reduce the nonlinear distortion caused by the power amplifier. Compared with the No-DPD case, both methods improve the NMSE, EVM, and ACPR results for all tested IBO values.

Polynomial DPD gives the best overall NMSE and EVM performance when it is trained and tested at the same IBO value. This is reasonable because the c_3 coefficient is selected specifically for that operating point. As a result, the predistorter matches the PA behavior more closely.

LUT-based DPD also provides clear improvement compared with the No-DPD case. Its EVM and NMSE results are slightly worse than Polynomial DPD in most cases, but its ACPR performance is close. In the generalization test, LUT DPD performs slightly better at higher IBO values, which suggests that the amplitude-based correction table can be more tolerant to some operating-point changes.

The learned c_3 coefficient decreases as IBO increases. This means that less predistortion is needed when the PA operates closer to the linear region. At low IBO values, the amplifier is driven harder and stronger correction is required.

The generalization test also shows that a DPD model trained at one IBO value does not always work optimally at other IBO values. When the operating point changes, the PA behavior also changes, so the trained predistorter may no longer match the PA exactly.

Overall, Polynomial DPD is more effective when the operating condition is known and fixed, while LUT-based DPD provides a useful alternative with comparable performance and slightly better robustness in some shifted conditions.

6. Limitations and Future Work

This study uses a simplified memoryless PA model. Therefore, the output depends only on the current input sample. In practical RF power amplifiers, memory effects may also appear due to thermal behavior, bias circuits, and frequency-dependent components. For this reason, a memory polynomial PA and DPD model can be investigated in future work.

Another limitation is that the PA coefficients used in this project are real-valued. Because of this, the model mainly represents AM/AM distortion and does not explicitly include AM/PM distortion. In a more realistic Saleh-type complex PA model, the output phase would also change with input amplitude. In that case, the DPD model would need to compensate both amplitude and phase distortion.

In future work, adaptive DPD techniques can also be considered. Instead of using a fixed predistorter for each operating point, the DPD parameters can be updated when the input power or IBO condition changes. This would improve robustness under different operating conditions.

Finally, more challenging waveforms such as OFDM can be tested. Since OFDM has a high peak-to-average power ratio, it would drive the PA into nonlinear regions more frequently and provide a more realistic test case for DPD performance.

7. Conclusion

This project analyzed the effect of power amplifier nonlinearity and compared different digital predistortion methods. A pulse-shaped 16-QAM signal was passed through a memoryless nonlinear PA model and evaluated with No DPD, Polynomial DPD, and LUT-based DPD.

The results show that both DPD methods improve the PA output compared with the No-DPD case. Polynomial DPD gives the best NMSE and EVM performance when it is trained and tested at the same IBO value. LUT-based DPD also provides clear improvement and shows slightly better robustness in some shifted operating conditions.

Overall, the results confirm that digital predistortion is an effective method for reducing nonlinear distortion, improving modulation quality, and limiting spectral regrowth.

8. AI Usage Declaration

AI-assisted tools were used for language refinement, report organization, and debugging support during the project. The MATLAB simulations, parameter choices, generated results, and engineering interpretations were checked and finalized by the author.